

Muhammed Raneesh C M

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Research Objective

Graduate researcher in Information Security with demonstrated expertise in network protocol analysis, cryptography, and systems security. Seeking research opportunities to advance cybersecurity knowledge through rigorous analysis of emerging protocols, vulnerability assessment, and security system design. Committed to contributing to research in network security, distributed systems, and applied cryptography.

Education

Master of Technology in Computer Science and Engineering (Information Security) 2023–2025

National Institute of Technology, Calicut, Kerala, India

CGPA: 8.39/10.0 (First Class with Distinction)

Roll No: M231041CS

Bachelor of Technology in Computer Science and Engineering 2017–2021

APJ Abdul Kalam Technological University, Kerala, India

CGPA: 8.92/10.0(Honours)

Academic Coursework

Master of Technology (Information Security) - NIT Calicut

Core Security Courses:

- **Foundations of Information Security** (4 Credits) - Grade: B
- **Topics in Cryptography** (3 Credits) - Grade: B
- **Topics in Information Security** (3 Credits) - Grade: A
- **Systems Security** (3 Credits) - Grade: B

Computer Systems & Networks:

- **Computer Networking** (3 Credits) - Grade: A
- **Distributed Computing and Big Data** (4 Credits) - Grade: A
- **Software Systems Lab** (4 Credits) - Grade: A

Theoretical Foundations:

- **Mathematical Foundations of Computer Science** (4 Credits) - Grade: C
- **Algorithms and Complexity** (4 Credits) - Grade: D
- **Theoretical Foundations of Machine Learning** (4 Credits) - Grade: C

Database & Research:

- **Topics in Database Design** (3 Credits) - Grade: B
- **Research Methodology** (2 Credits) - Grade: A

Research Project:

- **Project Phase I** (2 Credits) - Grade: B
- **Project Phase II** (3 Credits) - Grade: B
- **Project Phase III** (15 Credits) - Grade: B
- **Project Phase IV** (15 Credits) - Grade: S (Excellent)

Academic Performance Summary:

- **Total Credits:** 76 (Professional Core: 55, Program Electives: 19, Institute Electives: 2)
- **Classification:** First Class with Distinction (CGPA above 7.5)
- **Strong Performance Areas:** Information Security, Networking, Distributed Systems, Research

Research Experience

Security Analysis of HTTP/3 Protocol

IEEE TIFS - Under Review

- **Research Domain:** Network Protocol Security
- **Methodology:** Systematic analysis of HTTP/3 and QUIC protocol specifications through examination of RFC 9000, 9001, 9002, and related standards
- **Technical Contribution:** Identified potential security vulnerabilities through formal protocol analysis and threat modeling

- **Implementation:** Developed proof-of-concept security explorations using Python aioquic library for protocol state manipulation
- **Publication Status:** Submitted to IEEE Transactions on Information Forensics and Security (Impact Factor: 7.231)
- **Research Tools:** Python, aioquic, Wireshark

Research Interests

- **Network Protocol Security:** HTTP/3, QUIC, TLS 1.3, Emerging internet protocols
- **Applied Cryptography:** Protocol-level cryptographic implementations, Cryptanalysis
- **Systems Security:** Vulnerability analysis, Formal verification, Secure system design
- **Distributed Systems Security:** Security in cloud computing, Microservices architectures
- **Web Security:** Browser security, Web application vulnerabilities, Client-server protocols

Technical Expertise

- **Security Analysis Tools:** Wireshark, Burp Suite, Protocol Analyzers, Vulnerability Scanners
- **Programming Languages:** Python (Advanced), C, JavaScript, C#, Dart
- **Research Methodologies:** Formal verification, threat modeling, protocol analysis, experimental security testing
- **Systems Administration:** Linux (Debian, Fedora, Arch), Docker, Network Configuration
- **Development Frameworks:** React.js, Node.js, Flutter, Unity 3D, MongoDB

Project Experience

No Due App

[GitHub](#) | [Design](#)

- Spearheaded a 4-member development team to build a MERN stack application streamlining the No Due clearance process, delivered within a tight 5-day timeline.
- Designed the complete UI/UX, developed the client-side application, and integrated it with back-end services.
- Revolutionized the manual clearance workflow into a streamlined and efficient digital process.

Augmented Reality Educational Application

[GitHub](#) | [Design](#)

- Engineered an AR-based learning platform for Kerala SSLC students utilizing Unity 3D and Google ARCore.
- Created a Flutter-based companion application for downloading notes.

Project Coordinator App

[Design](#)

- Collaborated in a 6-member team to develop a MERN stack application to coordinate final year student projects.
- Designed the complete UI/UX in Figma, focusing on intuitive navigation and user workflow.
- Contributed to front-end development using React.

Publications & Academic Contributions

- **Under Review:** "Security Analysis of HTTP/3 Protocol: Vulnerability Assessment and Mitigation Strategies" - IEEE Transactions on Information Forensics and Security (TIFS), 2024

Academic Achievements

- **Master's Degree:** First Class with Distinction (CGPA: 8.39/10.0)
- **Bachelor's Degree:** Honours Degree (CGPA: 8.92/10.0)
- **Academic Leadership:** Led multiple collaborative research and development projects

Professional Skills

- **Research Skills:** Literature review, Experimental design, Data analysis, Technical writing
- **Communication:** Academic presentation, Technical documentation, Peer collaboration

- **Project Management:** Research project coordination, Timeline management, Team leadership
- **Languages:** English (Fluent), Malayalam (Native), Hindi (Conversational)